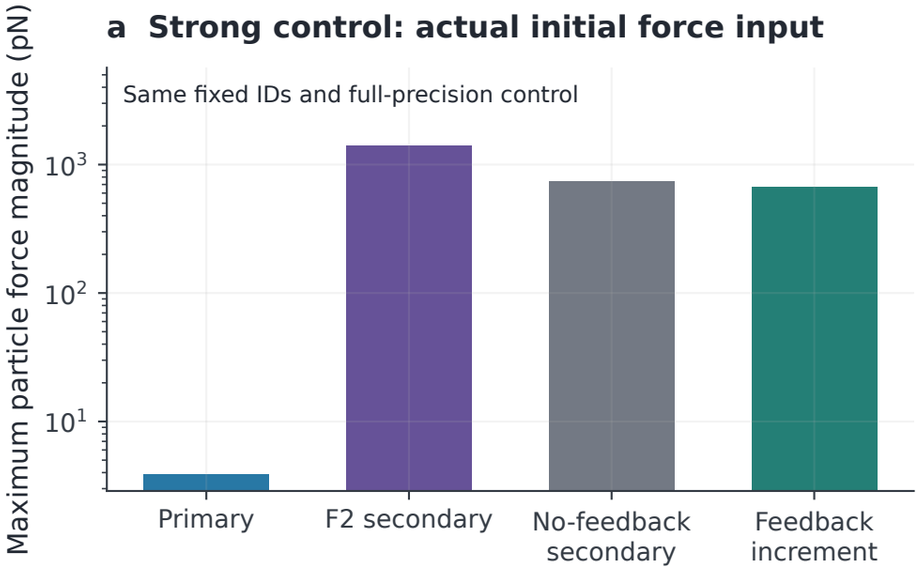
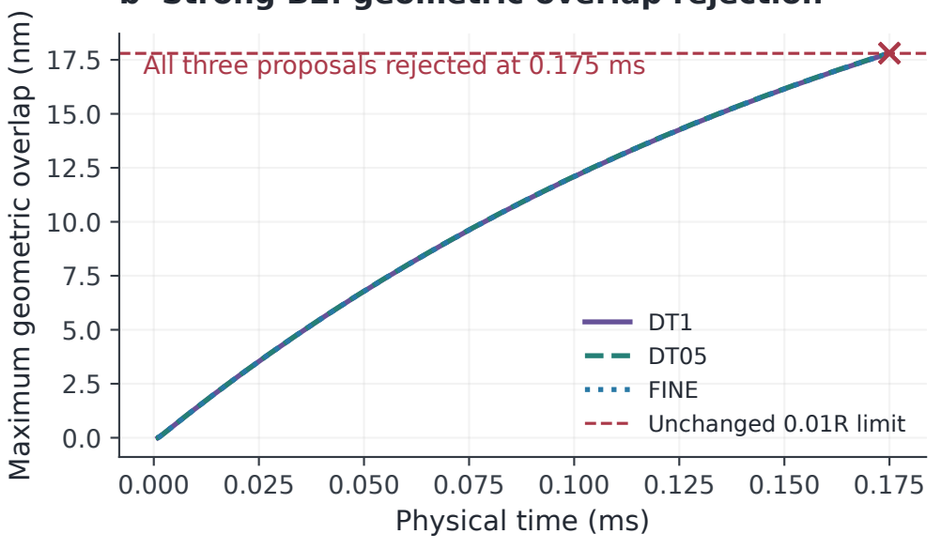
F10O | Strong-control contact audit

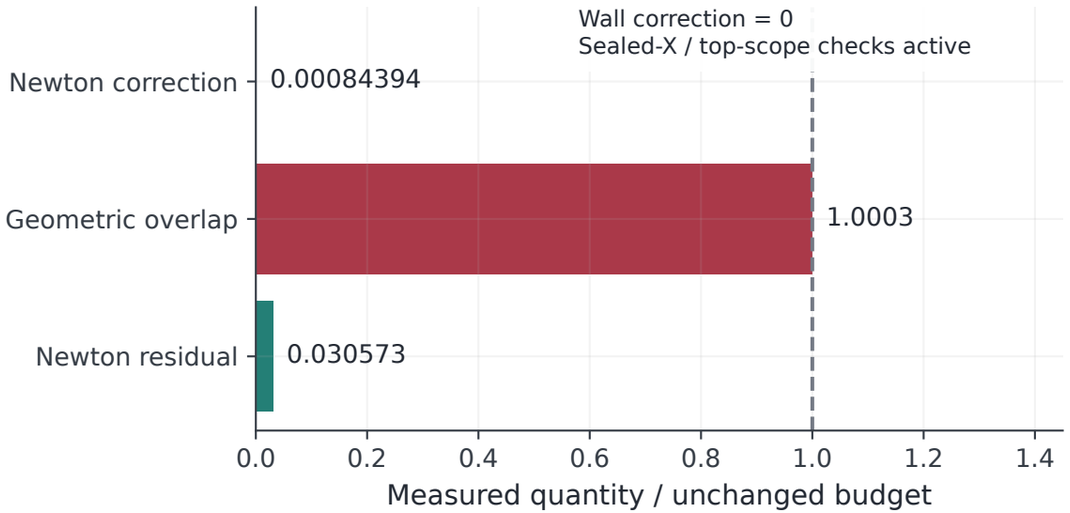
a Strong control: actual initial force input



b Strong B2: geometric overlap rejection



c Quality checks are independent



d Reused models, different contact approximations

M0 / PBD (historical F10N)
  Two local lubrication iterations; 12 geometric projection passes.
  Strong 2 ms same-ID dt difference: 6.416 µm.
  Cumulative PBD correction / predicted path: 0.971.

CPU B2 (unchanged F10H / F10K)
  Solve the same pair-normal resistance matrix.
  Implicit unilateral spring on actual final geometry.
  k = 0.04641319518378658 N/m; residual tolerance 10^{-13} m.
  No subsequent PBD; any wall correction rejects this interior test.

A converged soft-contact equation can still exceed the overlap limit.
M0–B2 trajectory differences are model contrasts, not pure time-integration error.